

Chapter 45: Preterm Birth — Revision Notes

Williams Obstetrics 26e, pp. 2036–2118 of the PDF. High-yield numbers are in **bold**. The ten tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 8 figures are included from the PDF.

Big picture: Preterm birth affects **~15 million births/yr worldwide** and is the **leading cause of death in children <5 years**. It is a *syndrome* with many pathways (not simply "labor that starts too soon"). Few preventive measures work well; the proven perinatal benefits come from **antenatal corticosteroids**, **antibiotics for PPROM** and **magnesium sulfate neuroprotection**.

1. Definitions

Term	Definition
Preterm birth	Delivery before 37 completed weeks (before 36 ⁶ / ₇ wk)
Late preterm (ACOG)	34–36 completed weeks
Early preterm (CDC)	Before 33⁶/₇ wk
WHO: extremely preterm	<28 wk
WHO: very preterm	28–32 wk
WHO: moderate-to-late preterm	32–37 wk

- These cutoffs lack a functional basis. **Prematurity** = incomplete organ development at birth (lungs → RDS).
- A preterm neonate can be SGA or LGA and still be preterm.

Birthweight category	Weight
Low birthweight (LBW)	1500–2500 g
Very low birthweight (VLBW)	1000–1500 g
Extremely low birthweight (ELBW)	<1000 g

2. Preterm Birth Rate Trends

- US rate **10.02% (2018) → 10.23% (2019)**. Fell 2007–2014, rising since.
 - The rise was mainly in **late-preterm births** (rising since 2013).
 - From **2014**, the NCHS uses the **obstetrical estimate** of gestational age instead of the LMP → current data are not directly comparable with older rates.
- National data do **not separate spontaneous from indicated** preterm births.
- **Racial disparity:** rates are markedly higher in **black women** than in white and Hispanic women. US rates exceed those of other industrialized countries; international rates are also rising.
- **Multifetal gestations** raise the rate → analyze singletons separately.

3. Preterm Newborn Morbidity and Mortality

- US 2018: **21,498 infant deaths; 66% were among preterm births**.
- Gestational age and morbidity/mortality are **inversely related**.
- Infant mortality at **<28 wk was 186x higher** than at 37–41 wk.

- Lower 5- and 10-min Apgar scores → higher relative risk of death in preterm neonates (may reflect immaturity rather than depression).
- **Survival reaches 95%** once birthweight is **≥1000 g**, or GA is **≥28 wk (females)** or **≥30 wk (males)** (Fanaroff 2007).

Table 45-1. Infant Mortality Rates in the United States in 2018

Gestational age (weeks)	Live Births	Infant Deaths
Total infants	3,791,712	21,498
<34	104,031	12,004
34–36	275,746	2263
<37	379,777	14,267
37–38	1,005,405	3135
39–41	2,292,705	3843
≥42	11,318	61

Table 45-2. Major Short- and Long-Term Problems in Very-Low-Birthweight Infants

Organ or System	Short-Term Problems	Long-Term Problems
Pulmonary	Respiratory distress syndrome, air leak, bronchopulmonary dysplasia, apnea of prematurity	Bronchopulmonary dysplasia, reactive airway disease, asthma
Gastrointestinal or nutritional	Hyperbilirubinemia, feeding intolerance, necrotizing enterocolitis, growth failure	Failure to thrive, short-bowel syndrome, cholestasis
Immunological	Hospital-acquired infection, immune deficiency, perinatal infection	Respiratory syncytial virus infection, bronchiolitis
Central nervous system	Intraventricular hemorrhage, periventricular leukomalacia, hydrocephalus	Cerebral palsy, hydrocephalus, cerebral atrophy, neurodevelopmental delay, hearing loss
Ophthalmological	Retinopathy of prematurity	Blindness, retinal detachment, myopia, strabismus
Cardiovascular	Hypotension, patent ductus arteriosus, pulmonary hypertension	Pulmonary hypertension, systemic hypertension, hypertension in adulthood
Renal	Water and electrolyte imbalance, acid-base disturbances	Hypertension in adulthood
Hematological	Iatrogenic anemia, need for frequent transfusions, anemia of prematurity	—
Endocrinological	Hypoglycemia, transiently low thyroxine levels, cortisol deficiency	Impaired glucose regulation, increased insulin resistance

Threshold of viability

- Fetuses **<500 g** once called "abortuses" are now live births (**5189** such US live births in 2019).
- **Threshold of viability** = lower limit of maturation compatible with extrauterine survival: currently **20⁰/₇** to **25⁶/₇** wk (the **perivable period**).
- Guidance: 2013 NICHD workshop (Raju 2014) → ACOG **Obstetric Care Consensus on Perivable Birth** (2019).

Perivable survival

- Birth **before 23 wk** usually ends in death: survival **~5–10%**, with significant morbidity among survivors.
- Survival at **24 wk** ranges **35–84%** internationally.
- Iowa (Watkins 2020), active management at 22–25 wk: **64%** of 22–23-wk survivors had mild or no neurodevelopmental impairment.

- **Ascertainment bias:** mean survival **45%** if the denominator is all live births vs **72%** if only NICU admissions.
- NICHD NRN, 22–24 wk (Younge 2017): survival **30%** → **36%** (2000–03 → 2008–11); survival without neurodevelopmental impairment **16%** → **20%**. At **22 wk only 1%** survived without impairment.

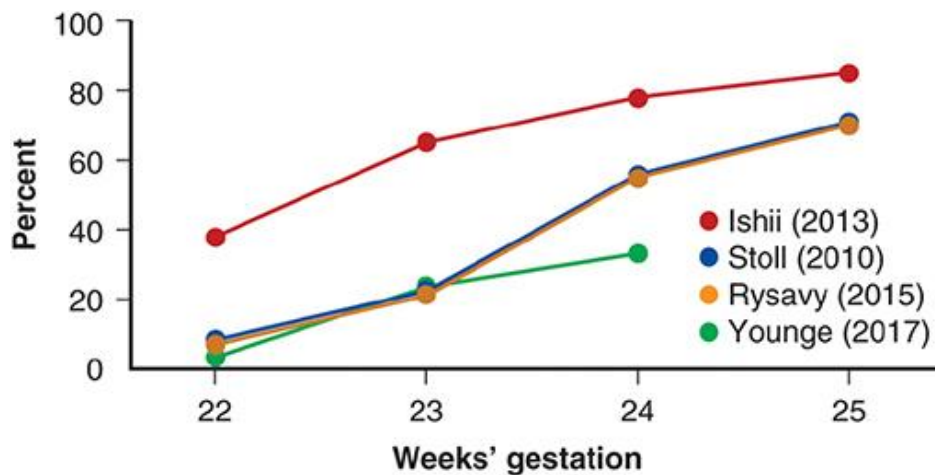


Figure 45-1. Neonatal survival at 22–25 weeks rises steeply with each week; curves differ by denominator (liveborn vs NICU vs intact survival).

Table 45-3. One-Year Survival Among Infants Delivered at 22–26 Weeks' Gestational Age in Sweden

Values are No./Total (%), for 2004–2007 → 2014–2016.

Outcome	22 wk 04–07	22 wk 14–16	23 wk 04–07	23 wk 14–16	24 wk 04–07	24 wk 14–16
All deliveries — Stillbirths	91/140 (65)	52/148 (35)	82/183 (45)	59/207 (29)	47/191 (25)	68/260 (26)
Live born	49/140 (35)	96/148 (65)	101/183 (55)	148/207 (71)	144/191 (75)	192/260 (74)
1-yr survival — NICU admission	5/17 (29)	29/50 (58)	53/81 (65)	91/138 (66)	96/132 (73)	151/191 (79)
Live born ^a	5/49 (10)	29/96 (30)	53/101 (52)	91/148 (61)	96/144 (67)	151/192 (79)
All deliveries ^b	5/140 (3.6)	29/148 (20)	53/183 (29)	91/207 (44)	96/191 (50)	151/260 (58)
1-yr survival, no major morbidity ^c — NICU admission	1/5 (20)	5/29 (17)	9/53 (17)	25/91 (28)	30/96 (31)	60/151 (40)
Live born ^d	1/49 (2.0)	5/96 (5.2)	9/101 (8.9)	25/148 (17)	30/144 (21)	60/192 (31)

Outcome	25 wk 04–07	25 wk 14–16	26 wk 04–07	26 wk 14–16	All 22–26 wk 04–07	All 22–26 wk 14–16
All deliveries — Stillbirths	45/250 (18)	58/277 (21)	39/245 (16)	36/304 (12)	304/1009 (30)	273/1196 (23)
Live born	205/250 (82)	219/277 (79)	206/245 (84)	268/304 (88)	705/1009 (70)	923/1196 (77)
1-yr survival — NICU admission	167/200 (84)	193/219 (88)	176/204 (86)	247/267 (93)	497/634 (78)	711/865 (82)
Live born ^a	167/205 (81)	193/219 (88)	176/206 (85)	247/268 (92)	497/705 (70)	711/923 (77)
All deliveries ^b	167/250 (67)	193/277 (70)	176/245 (72)	247/304 (81)	497/1009 (49)	711/1196 (59)
1-yr survival, no major morbidity ^c — NICU admission	75/167 (45)	104/193 (54)	111/176 (63)	161/247 (65)	226/497 (45)	355/711 (50)
Live born ^d	75/205 (37)	104/219 (47)	111/206 (54)	161/268 (60)	226/705 (32)	355/923 (38)

^aPrimary outcome. ^bIncludes stillbirths. ^cMajor neonatal morbidity = IVH grade 3 or 4; periventricular leukomalacia; necrotizing enterocolitis; retinopathy of prematurity stage 3, 4 or 5; or severe bronchopulmonary dysplasia.

^dSecondary outcome. NICU = neonatal intensive care unit.

Clinical management of threatened periviable birth

- **Nonmodifiable:** fetal gender, weight, plurality.
- **Potentially modifiable:** place of delivery, intent to intervene by cesarean, antenatal corticosteroids.
- Postnatal: initiation or withdrawal of intensive care.

Table 45-4. General Guidance for Threatened Periviable Birth

Intervention	<22 wk	22 ⁰ / ₇ –22 ⁶ / ₇ wk	23 ⁰ / ₇ –23 ⁶ / ₇ wk	24 ⁰ / ₇ wk +
Assess neonate for resuscitation	Not recommended	Consider	Consider	Recommended
Corticosteroid therapy	Not recommended	Not recommended	Consider	Recommended
Tocolysis to allow CS therapy	Not recommended	Not recommended	Consider	Recommended
MS for neuroprotection	Not recommended	Not recommended	Consider	Recommended
Antibiotics for PPROM	Consider	Consider	Consider	Recommended
GBS prophylaxis	Not recommended	Not recommended	Consider	Recommended
CD for fetal indication	Not recommended	Not recommended	Consider	Consider ^a /recommended

^aCD is considered for fetuses 24⁰/₇–24⁶/₇ wk but recommended for those 25⁰/₇ weeks and older. CD = cesarean delivery; CS = corticosteroid; GBS = group B streptococcus; MS = magnesium sulfate; PPROM = preterm prelabor rupture of membranes.

Memory hook: "22 = Consider, 23 = Consider all, 24 = Recommend all." Antibiotics for PPROM are the only thing considered even below 22 wk.

Cesarean at the limit of viability

- Controversial; observational data are inconsistent and cesarean adds maternal morbidity.
- Reddy 2012 (24–31⁶/₇ wk): **84%** of cephalic fetuses delivered vaginally with **no difference in neonatal mortality**. For **breech**, attempted vaginal delivery → **3x mortality**.
- Werner 2013 (24–34 wk): cesarean did **not** protect against death, IVH, seizures, RDS or subdural hemorrhage.

- Consensus: consider **cesarean for fetal indications at 23⁰/₇–24⁶/₇ wk; before 22 wk only for maternal indications.**
- Active management (e.g., corticosteroids) can be **uncoupled** from the decision for cesarean.
- Interventions at **22 vs 23 wk: 38.9% vs 78.3%** (Hajdu 2020). **~20%** of women have serious morbidity with periviable delivery.
- Individualized, multidisciplinary care; **perinatal palliative care** for life-limiting prematurity.

Late-preterm birth

- **34–36 wk = >70% of all preterm births** (72.9% in 2019).
- Parkland (McIntire 2008): 3% of births at 24–32 wk and **9% late preterm** → late preterm = **¼ of preterm births**.
- **~80%** follow idiopathic spontaneous preterm labor or PROM; **20%** other obstetrical complications.
- Higher neonatal mortality, morbidity and adverse neurodevelopment than term → prematurity care should include late preterm.

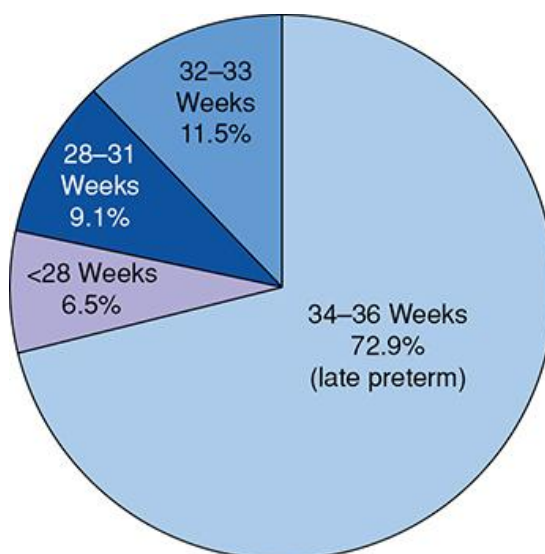


Figure 45-2. US preterm births 2019 by gestational age: 34–36 wk = 72.9%; 32–33 wk 11.5%; 28–31 wk 9.1%; <28 wk 6.5%.

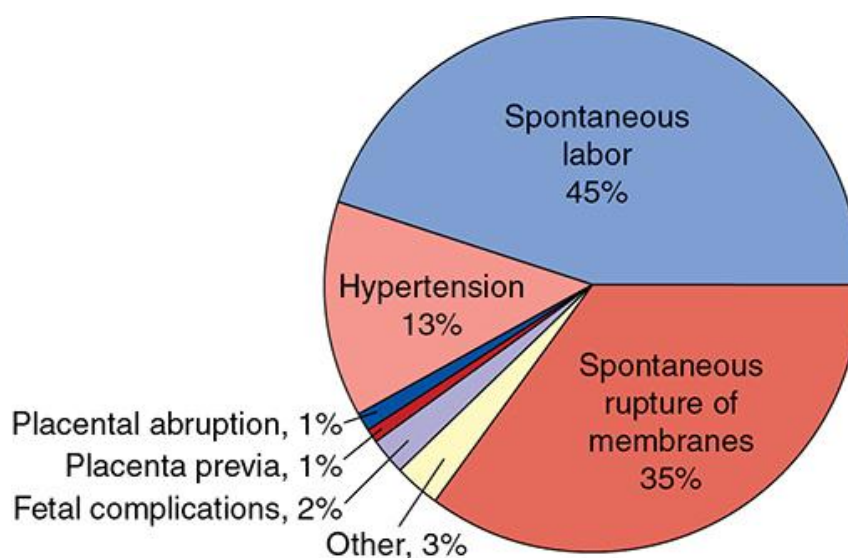


Figure 45-3. Causes of 21,771 late-preterm births at Parkland: spontaneous labor 45% + spontaneous membrane rupture 35% (= 80%); hypertension 13%.

Table 45-5. Neonatal Outcomes in Live Births Delivered Late Preterm Compared with Referent Group at 39 Weeks

Outcome	34 wk	35 wk	36 wk	39 wk
Neonatal death rate per 1000 live births	1.1	1.5	0.5	0.2
Morbidity (%)				
Resp. distress — Ventilator	3.3 ^a	1.7 ^a	0.8 ^a	0.3
Resp. distress — TTN	2.4 ^a	1.6 ^a	1.1 ^a	0.4
IVH — Grades 1, 2	0.5 ^a	0.2 ^a	0.06 ^a	0.01
IVH — Grades 3, 4	0	0.02	0.01	0.004
Sepsis — Work-up	31 ^a	22 ^a	15 ^a	12
Sepsis — Proven	0.5 ^a	0.4 ^a	0.2 ^b	0.1
Phototherapy	6.1 ^a	3.5 ^a	2.0 ^a	1
NEC	0.09 ^a	0.02 ^b	0.01	0.001
5-min Apgar ≤3	0.1	0.2 ^a	0.09†	0.06
Intubation	1.4 ^a	0.8 ^b	0.6	0.6
≥1 of above	34 ^a	24 ^a	17 ^a	14

^a*p* <.001 vs 39-wk referent. ^b*p* <.05 vs 39-wk referent. IVH = intraventricular hemorrhage; NEC = necrotizing enterocolitis; TTN = transient tachypnea. †Printed as "0.9", which would exceed the 34- and 35-wk values yet carries no significance marker; almost certainly 0.09.

4. Causes of Preterm Birth

Four direct causes:

1. Spontaneous unexplained preterm labor with intact membranes — **40–45%**
2. Idiopathic preterm prelabor rupture of membranes (PPROM) — **30–35%**
3. Delivery for maternal or fetal indications — **30–35%**
4. Twins and higher-order multiples: **>98% of triplets** and **60% of twins** are born preterm.
 - Preterm labor = several disease processes activating the **common parturition pathway**; a heterogeneous syndrome (genes, microbes, environment). No reliable predictive biomarker yet despite multiomics.

Spontaneous preterm labor

- Associations: multifetal pregnancy, intrauterine infection, bleeding, placental infarction, premature cervical dilation, cervical insufficiency, hydramnios, uterine fundal abnormalities, fetal anomalies; severe maternal illness (infection, autoimmune disease, gestational hypertension).
- Labor is the **final step** of changes that may start **days or weeks** earlier.
- **Four major mechanisms**: uterine overdistention, premature cervical change, infection, maternal–fetal stress.

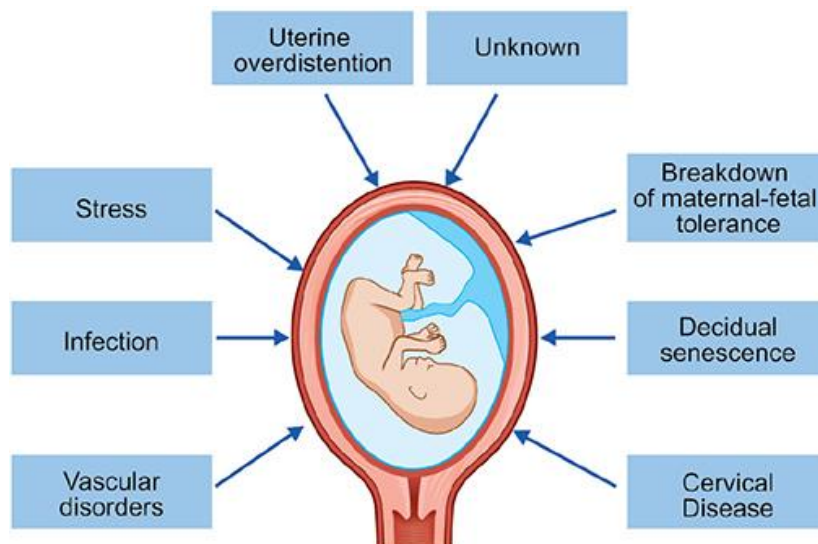


Figure 45-4. Pathways to spontaneous preterm labor: overdistention, stress, infection, vascular disorders, decidual senescence, breakdown of maternal–fetal tolerance, cervical disease, unknown.

Uterine overdistention (multifetal pregnancy, hydramnios)

- Myometrial stretch → **contraction-associated proteins** (gap-junction proteins, **oxytocin receptors**, prostaglandin synthase) and an "inflammatory pulse".
- Early activation of the **placental–fetal endocrine cascade** → inflammatory activation → contractions.

Cervical dysfunction

- An intact cervical **epithelial barrier** prevents ascending infection. Loss of **hyaluronan** or **GBS colonization** ↑ risk (GBS makes **hyaluronidase**).
- ↓ Mechanical competence: mutations in collagen/elastic fiber components → cervical insufficiency.

Infection

- **Only intraamniotic infection has been causally linked** to spontaneous preterm delivery.
- Routes: (1) transplacental, (2) via fallopian tubes, (3) **ascending from vagina/cervix — most common**.
- Colonization found in **25–40%** of preterm deliveries.
- **The earlier the preterm labor, the more likely infection.**
- Paradox: culture-positive amniotic fluid is as common at term labor (bacteria enter *because of* labor) as in preterm labor (bacteria *cause* it).
- **Sterile intraamniotic inflammation** (no detectable microbes) is also a risk factor.

Inflammatory cascade

- LPS → **toll-like receptors** (on phagocytes, decidua, cervical epithelium, trophoblast) → **IL-8** and **IL-1β**.
- IL-1β → ↑ IL-6, IL-8, TNF-α; leukocyte recruitment; matrix changes; fever/acute-phase response.
- IL-1β → **prostaglandins** → cervical ripening and loss of myometrial quiescence. COX-2 inhibition prevents inflammation-mediated preterm labor in mice.
- **MMPs** break down collagen/elastic fibers → weakens membranes and cervix.
- Amniotic fluid cytokines come mostly from recruited **mononuclear phagocytes and neutrophils**.

Microbiota

- The pregnant vaginal microbiome becomes **less diverse and more stable**.
- *Gardnerella vaginalis* and *Ureaplasma urealyticum* ↑ in some studies of preterm birth; asymptomatic *Candida* colonization is **not** linked.
- **Most consistent finding: *Lactobacillus crispatus*-dominant flora is protective.**
- The **placenta has no relevant microbiome** (de Goffau 2019: bacteria reflect labor or contamination, except *Streptococcus agalactiae*).

Preterm prelabor rupture of membranes (PPROM)

- **Definition:** spontaneous membrane rupture **before 37 completed weeks and before labor onset** (formerly "premature rupture").
- Major predisposing events: **intrauterine infection, oxidative-stress DNA damage, premature cellular senescence**.
- Risk factors: low socioeconomic status, **low BMI**, nutritional deficiency, **smoking, prior PPRM** (recurrence). Often none.
- Amniotic fluid bacteria in **~1/3** of PPRM cases.

Molecular changes

- More amnion **apoptosis** (endotoxin, IL-1 β , TNF- α).
- Oxidative stress $\rightarrow$ DNA damage $\rightarrow$ senescence $\rightarrow$ inflammation and proteolysis.
- Rupture $\rightarrow$ $\uparrow$ **thrombin** $\rightarrow$ MMPs and prostaglandins.
- Endotoxin/TNF- α $\rightarrow$ amnion releases **fFN** $\rightarrow$ binds **TLR4** in mesenchymal cells $\rightarrow$ $\uparrow$ **PGE₂** and **MMPs** $\rightarrow$ ripening, contractions, collagen breakdown.
- **Galectins** may mediate maternal–fetal tolerance. Preterm labor and PPRM share **inflammation** as the common mechanism.

Maternal–fetal stress

- Psychosocial stress (e.g., **racial discrimination**, childhood stress, depression, PTSD, **intimate partner violence**) is linked to preterm birth and LBW.
- Mechanisms: premature activation of the **placental–adrenal endocrine axis**; premature **cellular senescence**.

5. Contributing Factors

Association vs causation is hard to untangle.

Table 45-6. Factors That May Contribute to the Genesis of Preterm Labor

Factor	Examples
Prior preterm birth	Spontaneous, indicated
Lifestyle	Smoking, poor diet, illicit drug use, stress, heavy physical activity
Genetic	Recurrent, familial, ethnoracial, immunoregulatory genes
Infection	<i>Mycoplasma</i> spp., bacterial vaginosis
Periodontal disease	Gingivitis
Interpregnancy interval	Long or short

Prior preterm birth — the MOST important risk factor

- Parkland (Bloom 2001, ~16,000 women): preterm first birth $\rightarrow$ **3 $\times$ recurrence risk**.
- After **two** preterm births, **>1/3** have a third preterm birth.
- **70%** of recurrences occur **within 2 weeks** of the prior preterm GA; causes also recur.
- But women with prior preterm birth account for only **10%** of preterm births $\rightarrow$ **90% are not predictable** from history.
- Prior **indicated** preterm birth also predicts later **spontaneous** preterm birth.
- Recurrence depends on the **number, GA (severity)** and **order** of prior preterm births: preterm $\rightarrow$ term carries a lower risk than term $\rightarrow$ preterm.

Pregnancy factors

- **Threatened abortion** (bleeding at 6–13 wk, light or heavy) → ↑ preterm labor, abruption and loss before 24 wk.
- **Birth defects** → preterm birth and LBW.
- **Multifetal gestation** — preterm delivery is the main cause of its excess perinatal morbidity/mortality.

Lifestyle factors

- **Underweight and obese** mothers, young or advanced age, poverty, short stature, **vitamin C deficiency**, smoking, inadequate weight gain, illicit drugs.
- Long hours, **fixed night shifts**, hard physical labor → probably ↑ risk.
- **Aerobic exercise is safe**; leisure-time activity may ↓ risk. Work accommodations are the best way to protect pay and job.

Genetic factors

- Familial recurrence. GWAS variants in **EBF1, EEFSEC, AGTR2** (European-ancestry cohorts only).

Periodontal disease

- Gingivitis affects **up to 50%** of pregnant women; associated with preterm birth.
- **Treating it in pregnancy does NOT lower preterm birth** (Michalowicz 2006, 813 women).

Interpregnancy interval

- Intervals **<18 months and >59 months** → ↑ preterm birth and SGA. Causality questioned; the effect is modified by whether the prior birth was preterm.

Infection

- **Antimicrobial prophylaxis does not work:**
 - Interconceptional azithromycin + metronidazole every 4 months → no reduction (and possibly harmful).
 - Metronidazole + erythromycin at 20–24 wk (2661 women) → no reduction in preterm birth or histological chorioamnionitis.
 - Not recommended with preterm labor and intact membranes.
- **Bacterial vaginosis:** H₂O₂-producing lactobacilli replaced by anaerobes. Linked to abortion, preterm labor, PROM, chorioamnionitis, amniotic fluid infection — but **screening and treatment do NOT prevent preterm birth**, and treatment breeds resistance.

COVID-19

- Conflicting population data. Parkland: **no increase** in preterm birth (252 infected vs 3122 negative). Severe infection → **indicated** preterm delivery.

6. Diagnosis

Symptoms

- Contractions, pelvic pressure, menstrual-like cramps, watery discharge, low back pain — also common in normal pregnancy (Braxton Hicks).
- **ACOG definition of preterm labor: regular contractions + cervical change** (dilation, effacement or both), **or** regular contractions with **cervical dilation ≥2 cm** at presentation.
- Parkland (Chao 2011, 843 women, 24–33⁶/₇ wk, cervix <2 cm, sent home as false labor):
 - Birth <34 wk **2% vs 1%** in the general population (similar); **34–36 wk 5% vs 2%** (higher).
 - Discharged at **1 cm** → birth <34 wk **5% vs 1%**; **~90%** delivered within **21 days**.

Cervical change

- Cervical length/diameter at 18–30 wk are **identical in nulliparas and multiparas** → multiparity does not explain early dilation.
- Asymptomatic dilation **2–3 cm at 26–30 wk** → **~25%** deliver before 34 wk.
- Routine digital cervical exams at every visit **do not change outcomes** (Buekens 1994) — neither beneficial nor harmful.

Ambulatory uterine monitoring

- Home tocodynamometry (FDA-approved 1985) **does not reduce preterm birth** → **discouraged**.

Biomarkers

- **Fetal fibronectin (fFN)**: glycoprotein for implantation/placental adherence; reflects cervical stromal remodeling.
 - **Positive = >50 ng/mL** (ELISA). Avoid contamination with amniotic fluid or blood.
 - **ACOG does NOT recommend fFN screening**; fFN-based interventions in asymptomatic women have not improved outcomes.
- **pHlGFBP-1** and **PAMG-1**: **PAMG-1 predicts better than fFN**, especially in symptomatic women.
- Proteomic markers and microRNAs: promising, not yet recommended.

Cervical length measurement

- Progressively **shorter cervix** → **higher preterm birth rate** (Iams 1996).
- **Transvaginal** sonography is unaffected by obesity, cervical position or shadowing; done **after 16 wk**; **singletons only** (not multifetal outside research). Specific training advised (SMFM).
- **Prior spontaneous preterm birth** → **TVS cervical length screening recommended** (SMFM and ACOG).
- **No prior preterm birth**: universal screening "reasonable" but **debated / not mandatory**.
- nuMoM2b (Esplin 2017, 9410 nulliparas): cervical length + fFN had **poor predictive performance** → not routinely recommended in low-risk women. Cost concerns (Bloom and Leveno 2017).

7. Preterm Birth Prevention

- Dietary supplements show **no benefit**; prevention may be achievable in selected groups.

Cervical cerclage

Three indications (singletons):

1. **History-indicated** — recurrent second-trimester loss with **cervical insufficiency** (Ch 11).
2. **Ultrasound-indicated** — short cervix on sonography.
3. **Rescue / emergency** — cervical incompetence found with threatened preterm labor.
 - **Short cervix without prior preterm birth** → **cerclage offers NO benefit**. To 2004: screened 47,123; 253 with cervix **<15 mm** randomized → no difference in delivery <33 wk.
 - **Short cervix + prior preterm birth** → **may benefit**. Owen 2009 (302 women, cervix **<25 mm**): primary outcome negative, but with **<15 mm**, delivery <35 wk **30% vs 65%**.
 - Berghella 2011 IPD metaanalysis: cerclage ↓ preterm birth and composite perinatal morbidity/mortality with **prior spontaneous PTB + singleton + CL <25 mm**.
 - Caveat: trials counted losses at **16–17 wk** as preterm births — hard to separate from cervical incompetence.
 - **ACOG**: singleton + **prior spontaneous PTB + CL <25 mm + GA <24 wk** → **cerclage may be considered (vs vaginal progesterone)**.

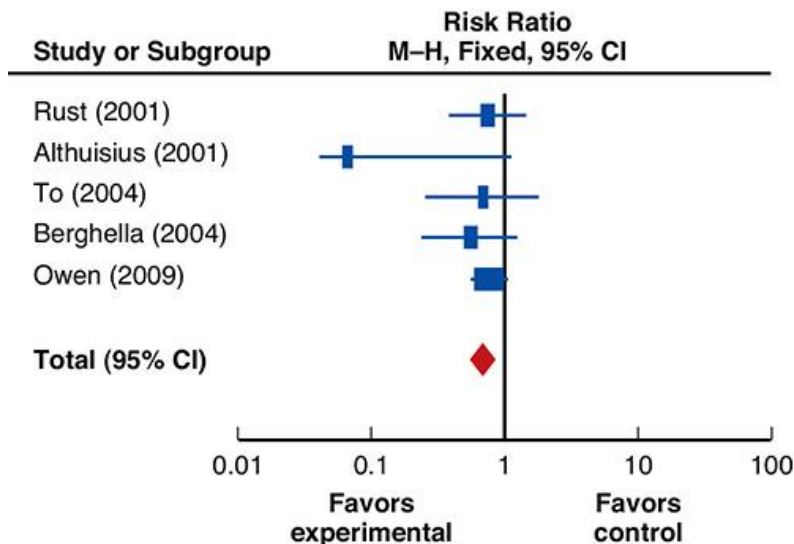


Figure 45-5. IPD metaanalysis: cerclage favored for prior preterm birth + cervical length <25 mm (pooled RR <1 for composite perinatal morbidity/mortality).

Technique trends

- Indication may guide suture choice; perioperative **cefazolin + indomethacin** with exam-indicated cerclage prolonged latency in one retrospective series (not adopted at Parkland).
- **After a failed vaginal cerclage → transabdominal cerclage is superior.**

Progestogen prophylaxis

- Rationale: human parturition involves **functional progesterone withdrawal** (↓ receptor activity), not a fall in levels.
- Benefit is **limited to singletons**; **no benefit in multifetal gestations**.
- ACOG and SMFM support progestogens in select singletons: **prior preterm birth OR no prior preterm birth but a sonographic short cervix**.

17-OHPC (17-alpha-hydroxyprogesterone caproate) for prior preterm birth

- The **only FDA-approved** drug for recurrent preterm birth (accelerated approval **2011**) — now disputed.
- **MFU trial (Meis 2003)**: 463 women, **250 mg IM weekly from 16 to 36 wk** → recurrence **36% vs 55%** (placebo).
 - Criticized: unexpectedly high placebo rate; **41% of placebo vs 28%** of 17-OHPC arm had ≥2 prior preterm births; dose chosen empirically.
- **Pricing**: Makena **\$1500/injection** (>\$30,000/pregnancy). Parkland used compounded 17-OHPC at **\$25/dose**: **ineffective** vs historical cohort (Nelson 2017, 430 women); plasma levels did not differ by outcome.
- **Pharmacology**: metabolized by **CYP3A**; **not** converted to 17α-hydroxyprogesterone; receptor affinity **~30%** of progesterone; half-life **median 16.2 days**; crosses the placenta (in cord plasma **44 days** after last dose). **Safe for the fetus** — no genital or developmental abnormalities at 48 months.
- **PROLONG trial** (Blackwell 2020; 1708 women, 2:1): **no reduction** in PTB <35 wk or neonatal composite (Table 45-7).
- **FDA**: Advisory Committee (Oct 2019) recommended **withdrawal**; CDER proposed withdrawal (Nov 2020); accelerated approval not yet withdrawn at time of writing.
- **ACOG and SMFM still endorse** 17-OHPC for recurrent PTB in singletons. EPPPIC 2021 metaanalysis: vaginal progesterone and 17-OHPC ↓ birth <34 wk, but **17-OHPC not statistically significant**, and trials of differing risk were pooled.

Table 45-7. Perinatal Outcomes of PROLONG Trial

Outcomes	17-OHPC (n = 1130)	Placebo (n = 578)	Relative Risk
Number assessed	1,113	574	
PTB <35⁰/₇ wk	122 (11.0)	66 (11.5)	0.95
— Spontaneous	93 (8.4)	51 (8.9)	0.93
— Indicated	28 (2.5)	14 (2.4)	1.03
Number assessed	1,112	572	
PTB <37⁰/₇ wk	257 (23.1)	125 (21.9)	1.06
— Spontaneous	209 (18.8)	98 (17.1)	1.10
— Indicated	46 (4.1)	26 (4.5)	0.91
Number assessed	1,116	574	
PTB <32⁰/₇ wk	54 (4.8)	30 (5.2)	0.92
— Spontaneous	38 (3.4)	22 (3.8)	0.88
— Indicated	15 (1.3)	7 (1.2)	1.11
Composite M&M	61 (5.6)	28 (5.0)	1.12
Death	6 (0.5)	3 (0.5)	0.98
BPD	6 (0.5)	1 (0.2)	3.02
RDS	54 (4.9)	26 (4.7)	1.06
NEC	2 (0.2)	2 (0.4)	0.5
IVH, grade 3 or 4	2 (0.2)	1 (0.2)	0.99
Sepsis	5 (0.5)	3 (0.5)	0.84

Values are n (%). The printed table places each RR on the "Number assessed" line; it is moved here to the PTB row it belongs to. 17-OHPC = 17-hydroxyprogesterone caproate; BPD = bronchopulmonary dysplasia; IVH = intraventricular hemorrhage; M&M = mortality and morbidity; NEC = necrotizing enterocolitis; PTB = preterm birth; RDS = respiratory distress syndrome.

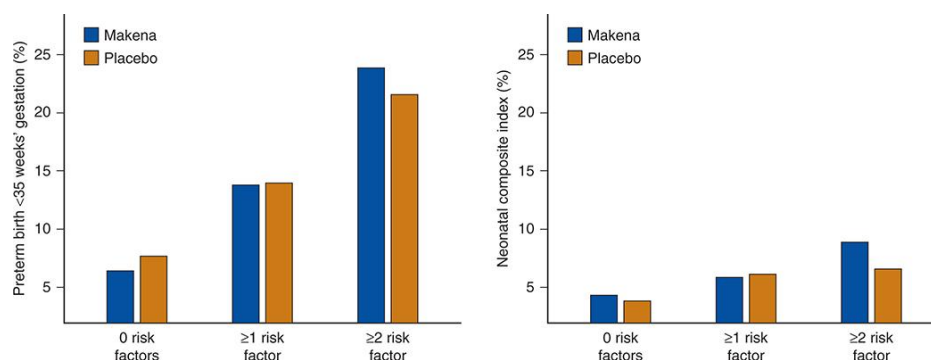


Figure 45-6. PROLONG post-hoc risk-profile analysis: Makena (17-OHPC) did not beat placebo for PTB <35 wk or neonatal composite, even with ≥2 risk factors.

Vaginal progesterone for a short cervix (no prior preterm birth)

- **Fonseca 2007:** CL ≤15 mm, **200-mg micronized progesterone vaginal capsules nightly, 24–34 wk** → ↓ spontaneous delivery <34 wk. (Mixed population incl. twins and prior PTB.)
- **Hassan 2011:** CL 10–20 mm, **90-mg vaginal gel daily** → ↓ PTB <33 wk.
- **Grobman 2012:** nulliparas, CL <30 mm, **17-OHPC IM** from 16–22³/₇ wk → **no benefit** at any cervical length.
- → **Vaginal progesterone, but not IM 17-OHPC, appears to benefit a short cervix.** Natural progesterone ≠ synthetic 17-OHPC; 17-OHPC has no local anti-inflammatory effect.

- **OPPTIMUM (Norman 2016)**: high-risk women (prior spontaneous birth ≤ 34 wk, CL ≤ 25 mm, or positive fFN + risk factors) → vaginal progesterone **did NOT** ↓ PTB, neonatal composite, or change 2-year cognitive scores (no harm either).
- ACOG: universal CL screening without prior PTB **not mandatory**; could be considered in the context of vaginal progesterone treatment.

Table 45-8. Randomized Trials of Progestogen Compounds Given Prophylactically to Prevent Preterm Labor

Investigator	Women Randomized	Cervical Length ^a	Progestogen Compound	Progestogen vs Placebo
Fonseca (2007)	n = 250; 5% nulliparous, 10% twins, 15% prior PTB; 8 hospitals: UK, Greece, Brazil, Chile	<15 mm	Progesterone, 200-mg vaginal capsules daily	Delivery <34 weeks: 19% vs 34% , p = .02
Hassan (2011)	n = 465; singletons only; 55% nulliparous; 13% prior PTB; 44 hospitals in 10 countries	10–20 mm	Progesterone, 90-mg vaginal gel daily	Delivery <33 weeks: 9% vs 16% , p = .02
Grobman (2012)	n = 657; singletons only; nulliparous only; 14 centers across US	<30 mm	17-OHPC, 250 mg IM weekly	Delivery <37 weeks: 25% vs 24% , p = NS

^aDetermined sonographically. 17-OHPC = 17-hydroxyprogesterone caproate; IM = intramuscularly; NS = nonsignificant; PTB = preterm birth. The text gives the Fonseca threshold as ≤ 15 mm; the table prints <15 mm.

Progestogens summary

- Evidence remains **conflicting**; nearly every indication can be challenged. A major review of progesterone use and a search for alternatives is warranted (Norman 2016).

Geography-based public health-care programs

- A well-organized prenatal system lowers preterm birth in high-risk indigent populations.
- **Parkland 1988–2006**: preterm rate fell (~10.5% → ~5%) as prenatal visit attendance rose — community clinics, nurse-practitioner protocols, central high-risk MFM clinics. Similar results at UAB.

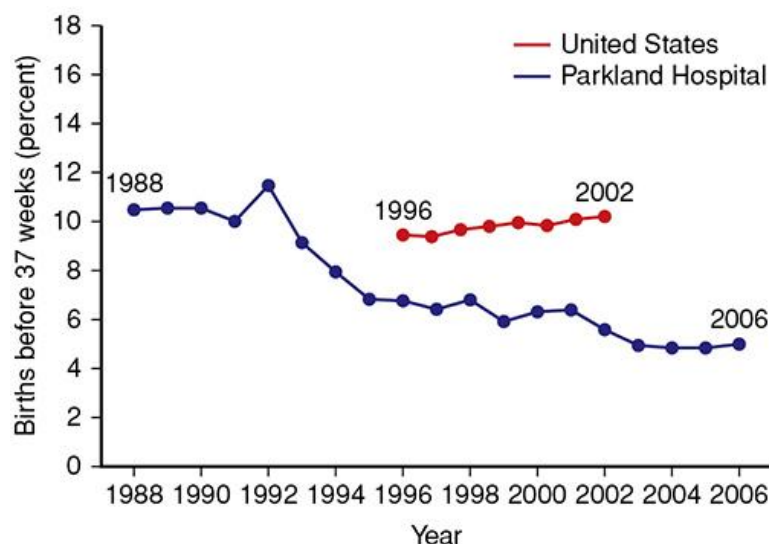


Figure 45-7. Parkland singleton births <37 wk fell from ~10.5% (1988) to ~5% (2006), while the US rate stayed ~9.5–10% (1996–2002).

Aspirin

- Low-dose aspirin **81 mg** from **6⁰/7–13⁶/7 wk** in 11,976 nulliparas (LMICs; Hoffman 2020) → ↓ birth <37 wk and ↓ perinatal mortality (did not separate spontaneous vs indicated).

- **ACOG does NOT recommend** aspirin to prevent spontaneous preterm birth **without preeclampsia risk factors**.

8. Management of Preterm Prelabor Rupture of Membranes

- History of fluid leakage (gush or stream) → **sterile speculum exam** for pooling or fluid from the os (Ch 22).
- **Sonography**: amniotic fluid volume, presenting part, gestational age.

Table 45-9. Management of Prelabor Rupture of Membranes

Gestational age	Management
37⁰/₇ weeks or older	Deliver. GBS prophylaxis as indicated (Table 67-3). Treat any intraamniotic infection.
34⁰/₇–36⁶/₇ weeks	Expectant management or deliver. Neonatal consultation. Consider single course of corticosteroids if: no prior corticosteroids; no chorioamnionitis; ≥24 hr anticipated from administration to delivery; <7 d anticipated from administration to delivery. For delivery, GBS prophylaxis as indicated; for expectant care, GBS screening. Treat any intraamniotic infection and deliver.
24⁰/₇–33⁶/₇ weeks	Expectant management , if no maternofetal indications favoring delivery. Neonatal and perinatal consultations. Single course of corticosteroids ; rescue course suitable but not supported by robust data. GBS screening on admission; for delivery, GBS prophylaxis as indicated. Antibiotics recommended to prolong latency. For pregnancies <32⁰/₇ weeks , magnesium sulfate for neuroprotection before anticipated delivery. Treat any intraamniotic infection and deliver.
Perivable	See Table 45-4

GBS = group B streptococcus. Source: ACOG 2019b; 2020g,h.

Natural history

- Latency is **inversely related to GA at rupture** — few days are gained in the third trimester.
- Parkland (Cox 1988, 24–34 wk): PPROM in **1.7%** of pregnancies; **76% already in labor**, 5% delivered for complications → only **19%** suitable for expectant care; delivery delayed ≥48 h in only **7%**. Neonatal deaths **0%** in that group vs **8%** if delivered within 48 h.

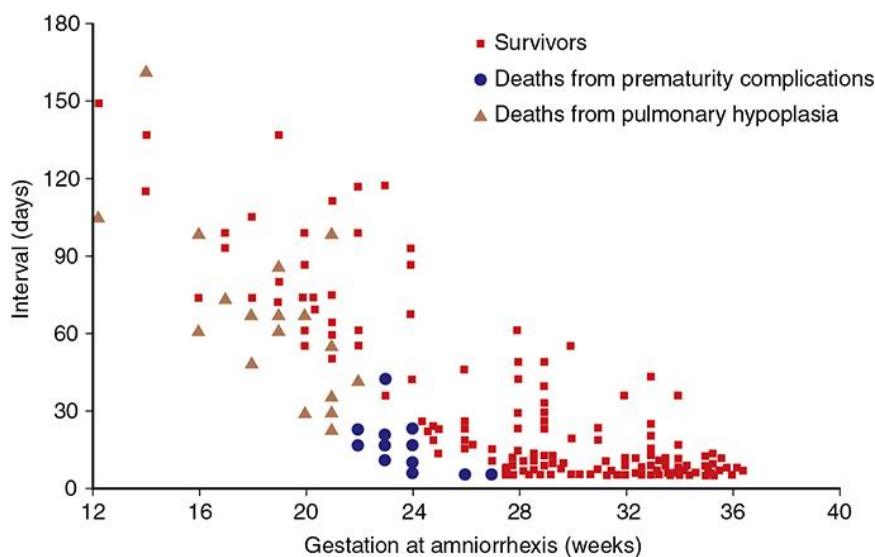


Figure 45-8. Rupture-to-delivery interval shortens as GA at rupture rises; deaths from pulmonary hypoplasia cluster with rupture before ~22–23 wk.

Hospitalization

- Most clinicians admit; most women labor **within a week**.

- Carlan 1993 (67 women): home care halved maternal stay (**14 vs 7 days**) with no difference found — too small to judge **cord prolapse** safety.
- Suggested surveillance: periodic sonography for growth, FHR monitoring, watch for infection (**fever**).
- Periviable PPROM: brief observation, then outpatient care in select cases; admit at viability.

Intentional delivery vs expectant management

- PPROM **<25 wk** managed expectantly (Morales 1993): mean **11 days** gained; **41%** alive at 1 yr but only **27%** neurologically normal.
- **Chorioamnionitis** is a recognized risk factor for neonatal neurological injury.
- Cochrane (Bond 2017; 12 RCTs, 3617 women): **no difference in neonatal sepsis**; early birth ↓ chorioamnionitis but ↑ prematurity sequelae → **expectant management with careful monitoring preferred <37 wk** if no contraindication.
- Late-preterm PROM: immediate delivery and expectant management give **comparable** composite neonatal outcomes.
- **ACOG:**
 - **24⁰/₇–33⁶/₇ wk** → **expectant management** unless nonreassuring fetal status, clinical chorioamnionitis or abruption.
 - **34⁰/₇–36⁶/₇ wk** → **expectant management OR delivery** (shared decision).
 - **Do not go beyond 37⁰/₇ wk.**
- Parkland favors **immediate delivery after 34 wk**.

Expectant management — specific issues

- **Digital cervical exam:** 1–2 exams shortened latency (**3 vs 5 days**) without worse outcomes → use judiciously.
- **PPROM after genetic amniocentesis:** much better outcome — perinatal survival **91%**; fluid usually normal again in **~2 weeks**; outpatient care.
- **Tocolysis:** prophylactic → no neonatal benefit and ↑ **chorioamnionitis**; therapeutic → no benefit.
- **Cerclage in place:** outcomes similar to controls; retention >24 h may prolong pregnancy but risks infection. Parkland retains it if no infection or labor.
- **Pulmonary hypoplasia:** up to **30%** of survivors after PPROM **<24 wk** → **23 wk** is a threshold. Residual fluid volume before 26 wk is prognostic. Oligohydramnios → **limb compression deformities**.
- **Active herpes:** expectant management — prematurity risk outweighs infection risk.
- **Noncephalic presentation** → ↑ **cord prolapse**, especially <26 wk.
- **Membrane sealants:** insufficient evidence; not used.

Clinical chorioamnionitis

- If diagnosed → **prompt delivery, preferably vaginal**.
- Maternal leukocytosis is inconsistent; **fever is the only reliable indicator**. Traditional threshold **≥38.0°C (100.4°F)** (still used at Parkland).
- NICHD 2015: "**Triple I**" = intraamniotic infection and inflammation.
- **ACOG (2019) definitions:**
 - **Suspected intraamniotic infection:** temperature **≥39.0°C**, OR **38.0–38.9°C + one additional clinical risk factor**.
 - Risk factors: low parity, multiple digital exams, internal uterine/fetal monitors, meconium-stained fluid, genital pathogens (**GBS**, STIs).
 - **Isolated maternal fever:** **38.0–38.9°C** with no additional risk factor.
- VLBW neonates of infected mothers (**~7%**): ↑ sepsis, RDS, early seizures, IVH, PVL. Intraamniotic infection ↑ **cerebral palsy**. Intrapartum fever (**1.6%**) predicts infection-related neonatal death.

Antimicrobial therapy

- Metaanalysis (Mercer 1995, PPROM <35 wk): ↓ chorioamnionitis, ↓ neonatal sepsis, more pregnancies prolonged **7 days**; survival and RDS unchanged.
- **MFMU trial (Mercer 1997), 24–32 wk: IV ampicillin + erythromycin q6h × 48 h → oral amoxicillin + erythromycin q8h × 5 days** (7 days total).
 - ↓ RDS, NEC and composite morbidity; undelivered at 7 days **50% vs 25%**; more undelivered at 14 and 21 days. Unaffected by GBS status.
- **3-day** vs 7-day ampicillin (or ampicillin-sulbactam) regimens appear equally effective.
- **Azithromycin can replace erythromycin** (better side effects).
- **⚠ Avoid amoxicillin-clavulanate → ↑ necrotizing enterocolitis** (ORACLE).
- Epoch comparison (Stoll 2002, 400–1500 g): overall early-onset sepsis unchanged; **GBS sepsis 5.9 → 1.7/1000**, but **E. coli sepsis 3.2 → 6.8/1000**, **~85% ampicillin-resistant**; coliform sepsis more lethal.
- No effect on child health at **7 years** (ORACLE).

Corticosteroids in PPROM

- **Single course at 24⁰/₇–34⁰/₇ wk** (ACOG). **May consider from 23⁰/₇ wk** if delivery risk within 7 days.
- 34–36⁶/₇ wk and rescue courses: see below.

9. Management of Preterm Labor with Intact Membranes

- Managed much like PPROM; aim to delay delivery before **34 wk** where possible.

Amniocentesis to detect infection

- **~1 in 10** women with preterm labor and intact membranes have (mostly subclinical) intraamniotic infection, but **routine amniocentesis offers little value**.

Antenatal corticosteroids

- Liggins and Howie **1972**: ↓ RDS and neonatal mortality if birth delayed **≥24 h** after betamethasone. No ill effects into adulthood.
- NIH Consensus **1995** recommended them; NIH **2000**: reasonable also in hypertension, diabetes, multifetal gestation, FGR, hydrops (insufficient data).
- Cochrane (Roberts 2017; 30 studies): ↓ **perinatal death, neonatal death, RDS, IVH, NEC, mechanical ventilation, systemic infection in first 48 h**. No benefit for chronic lung disease, childhood death or neurodevelopmental delay. **No ↑ chorioamnionitis**.
- LMIC cluster trial (Althabe 2015): ↑ perinatal mortality — attributed to **inaccurate dating**; WHO ACTION trial (**dexamethasone, 26⁰/₇–33⁶/₇ wk**) stopped early for **benefit** (↓ stillbirth or neonatal death).
- **ACOG: single course at 24–34 wk if at risk of delivery within 7 days**. May consider at **23 wk**; periviable use should match resuscitation plans.

Agent	Regimen
Betamethasone	12 mg IM × 2 doses, 24 h apart
Dexamethasone	6 mg IM q12h × 4 doses

- Equivalent efficacy for lung maturation and 2-year neurosensory outcome.
- **Give the first dose even if the course cannot be completed** (<24 h still helps).

Late-preterm steroids (ALPS; Gyamfi-Bannerman 2016)

- Betamethasone at **34⁰/₇–36⁶/₇ wk** → respiratory composite **11.6% vs 14.4%**. Only 60% of 2831 women got both doses.

- ↑ **Neonatal hypoglycemia** (possible developmental delay); the largest effect was on **TTN (6.7% vs 9.9%)**, a self-limited condition — and these TTN rates were 3–4× those of the Consortium on Safe Labor.
- ACOG and SMFM: **consider** a single course at 34⁰/₇–36⁶/₇ wk. **Parkland does not give steroids beyond 34 wk.**

Repeated courses

- Weekly repeat courses ↓ respiratory morbidity. Crowther 2007: no harm to 2 yr. Wapner 2007: **nonsignificant ↑ cerebral palsy** → "early gain, long-term questions." **Single course** recommended.

Rescue therapy

- = a second course when delivery is imminent and **>7 days** since the first.
- Peltoniemi 2007: single 12-mg rescue dose paradoxically ↑ RDS. Garite 2009 (<33 wk): ↓ respiratory complications and composite morbidity; mortality unchanged.
- **ACOG: consider one rescue course if <34 wk and the prior course was ≥7 days earlier.** Effects beyond 34 wk unknown. Parkland gives no additional courses.

Magnesium sulfate for neuroprotection

- Observational: VLBW neonates exposed to MgSO₄ (for preterm labor or preeclampsia) had less **cerebral palsy**.
- ACTOMgSO₄ (Crowther 2003, <30 wk, 1063 women): ↓ neonatal death and CP but underpowered. French trial similar.
- **BEAM (Rouse 2008)**: 2241 women, **24–31 wk**, imminent delivery.
 - **6-g IV bolus over 20–30 min, then 2 g/hr.**
 - Stopped after **12 h** if delivery not imminent; restarted when imminent, with a new loading dose if **>6 h** off.
 - Infusing at delivery in ~½. 2-year follow-up in **96%**.
- Interpretation: prevents CP at any GA (one view) vs only **<28 wk** (other view).
- Doyle 2009 metaanalysis (5 RCTs, 6145 infants): ↓ CP; **NNT = 63**. Criticisms: random error, no dose–response.
- **ACOG**: centers electing prophylaxis should develop **specific guidelines** (including PPRM).
- 22–26 wk: **steroids + MgSO₄** → less neurodevelopmental impairment or death than steroids alone.

Table 45-10. Magnesium Sulfate for the Prevention of Cerebral Palsy^a

Perinatal Outcome ^a	Magnesium Sulfate No. (%)	Placebo No. (%)	Relative Risk (95% CI)
Infants with 2-year follow-up	1041 (100)	1095 (100)	—
Fetal or infant death	99 (9.5)	93 (8.5)	1.12 (0.85–1.47)
Moderate or severe cerebral palsy — Overall	20/1041 (1.9)	38/1095 (3.5)†	0.55 (0.32–0.95)
— 24–27 weeks ^b	12/442 (2.7)	30/496 (6)	0.45 (0.23–0.87)
— 28–31 weeks ^b	8/599 (1.3)	8/599 (1.3)	1.00 (0.38–2.65)

^aSelected results from the Beneficial Effects of Antenatal Magnesium Sulfate (BEAM) Study. ^bWeeks' gestation at randomization. CI = confidence interval. †Printed typos corrected: placebo overall CP prints as "3/1095 (3.4)" (the subgroups sum to 38/1095 = 3.5%); the subgroup labels print as "<28–31" and "≥24–27" — the numbers fit 442 + 599 = 1041 randomized at 24–27 and 28–31 wk respectively, and the text states the benefit was seen before 28 wk; "confidence index" should read "confidence interval".

Antimicrobials for preterm labor with intact membranes

- **Not recommended**: no reduction in preterm birth (Cochrane, Flenady 2013); more short- and long-term harm.
- **ORACLE II** (6295 women): no benefit; ↑ **cerebral palsy at 7 years** with antibiotic exposure.
- **Clindamycin for BV** does not lower preterm birth.
- Distinct from **GBS prophylaxis**, which should be given.

Bed rest

- Widely prescribed, **unsupported**. Harms: **thromboembolism, deconditioning, bone loss**. Activity restriction → **2.5×** PTB <34 wk (likely ascertainment bias).
- **SMFM and ACOG recommend against activity restriction.**

Cervical pessaries

- Arabin silicone pessary: Goya 2012 (CL ≤25 mm) → spontaneous birth <34 wk **6% vs 27%**; later trials negative.
- Systematic review (4687 women): **no benefit** in singletons or twins with short cervix, or unselected twins.
SMFM: research protocols only.

Emergency (rescue) cerclage

- **Contraindicated with labor and active cervical change.**
- Althuisius 2003 (<27 wk, 23 women): emergency McDonald cerclage + bed rest delayed delivery **54 vs 24 days**.
- Poor-prognosis factors: **nulliparity, membranes prolapsing beyond the external os, cerclage before 22 wk**.
- Twins with 1–5 cm dilation at 16–23⁶/₇ wk (Roman 2020, all given indomethacin + antibiotics): ↓ PTB at all GAs; birth <28 wk ↓ **50%**.
- Reasonable with counseling for midgestation cervical dilation; unclear whether it truly helps or just ↑ rupture/infection.

Tocolysis

- **No tocolytic is completely effective.** They may delay delivery up to **48 hours** → time for **corticosteroids** and **transfer** to a higher-level center.
- No direct neonatal benefit shown; **maintenance tocolysis is ineffective**.
- Recommended short-term agents: **β-agonists, MgSO₄, calcium-channel blockers, indomethacin**. Most do not use them **after 33 wk**.

Agent	Mechanism	Evidence	Key adverse effects / cautions
β-agonists (ritodrine — withdrawn 2003; terbutaline)	↓ Intracellular Ca ²⁺	Delay 24–48 h ; oral and SC-pump terbutaline ineffective	Pulmonary edema (Na/water retention; worse with multifetal, steroids, >24 h tocolysis, large IV crystalloid), arrhythmia, myocardial ischemia. FDA 2011 warning → short-term inpatient use only ; 0.25 mg SC for uterine tachysystole
Magnesium sulfate	Calcium antagonist	Ineffective vs saline (Cox 1990); possibly harmful	Pulmonary edema (rare at Parkland). FDA 2013: avoid >5–7 days → fetal bone thinning/fractures (low fetal Ca)
Indomethacin (nonselective COX inhibitor)	↓ Prostaglandin synthesis	Not better than ritodrine or MgSO ₄ ; ineffective for short cervix	Oligohydramnios (reversible) → limit to 24–48 h ; debated NEC and premature ductus closure
Nifedipine (CCB)	Blocks Ca ²⁺ entry	Safer and more effective than β-agonists ; = MgSO ₄ ; = atosiban; Parkland RCT: no effect on PTB (stopped for futility)	⚠ Nifedipine + MgSO₄ is dangerous — enhanced neuromuscular blockade (cardiorespiratory compromise)
Atosiban (oxytocin-receptor antagonist)	Oxytocin antagonism	Used in Europe; no neonatal benefit, linked with neonatal morbidity	Not FDA-approved
Nitric oxide donors (nitroglycerin)	Smooth-muscle relaxation	Ineffective / not superior	Maternal hypotension

Labor and delivery

- **Continuous electronic FHR monitoring** preferred. **Fetal tachycardia** (esp. with ruptured membranes) suggests **sepsis**.
- Intrapartum **acidemia** (umbilical artery **pH <7.0**) worsens prematurity complications.
- Give **GBS prophylaxis** (Ch 67).
- **No routine episiotomy or forceps** to "protect" the preterm head.
- Skilled resuscitation team present; **delivery in tertiary centers** improves survival; team simulation helps community hospitals.
- Cesarean does **not** prevent germinal matrix/IVH → **cesarean only for usual obstetrical indications**.

Quick-Recall: What Works and What Doesn't

Proven / Recommended	Consider / Debated	Not effective / Not recommended
Corticosteroids 24–34 wk (single course)	Steroids at 23 wk; late-preterm (34–36 ^{6/7}) steroids; one rescue course <34 wk	Repeated weekly courses
Antibiotics for PPROM (ampicillin + erythromycin/azithromycin)	Cerclage retention after PPROM	Amoxicillin-clavulanate (NEC)
MgSO₄ neuroprotection (<32 wk; guideline-based)	Expectant vs delivery at 34–36 ^{6/7} wk PPROM	Antibiotics for preterm labor with intact membranes (↑ CP at 7 yr)
Cervical length screening with prior spontaneous PTB	Universal CL screening without prior PTB	fFN screening; home uterine monitoring; routine digital exams
Vaginal progesterone for short cervix (Fonseca, Hassan)	17-OHPC for prior PTB (ACOG/SMFM endorse; FDA proposed withdrawal)	17-OHPC for short cervix; progestogens in twins
Cerclage: prior PTB + CL <25 mm + <24 wk	Rescue cerclage; transabdominal cerclage after failed cerclage	Cerclage for short cervix alone ; pessary (research only)
Tocolysis ≤48 h for steroids/transfer	Indomethacin with cerclage	Maintenance tocolysis; bed rest; BV or periodontal treatment; aspirin without preeclampsia risk factors
GBS prophylaxis; continuous FHR monitoring	Cesarean for fetal indication at 23–24 wk	Routine cesarean, episiotomy or forceps for prematurity

Key Numbers to Memorize

- Preterm **<37 wk**; late preterm **34–36^{6/7} wk** = **>70%** of preterm births; US rate **~10.2%** (2019)
- LBW **1500–2500 g**; VLBW **1000–1500 g**; ELBW **<1000 g**
- Viability window **20^{0/7}–25^{6/7} wk**; survival <23 wk **5–10%**; 95% survival at **≥1000 g / ≥28 wk (F) / ≥30 wk (M)**
- Causes: spontaneous labor **40–45%**, PPROM **30–35%**, indicated **30–35%**; **>98%** triplets / **60%** twins preterm
- Prior PTB → **3x** recurrence; **70%** recur within **2 wk** of prior GA; but only **10%** of all PTB
- Preterm labor = regular contractions + cervical change, or **≥2 cm** dilation
- fFN positive **>50 ng/mL**; short cervix **<25 mm**; TVS after **16 wk**
- Interpregnancy interval **<18 or >59 months**
- 17-OHPC **250 mg IM weekly, 16–36 wk** (36% vs 55%); half-life **16.2 days**; PROLONG RR **0.95**
- Vaginal progesterone **200 mg capsule** or **90 mg gel** daily
- Betamethasone **12 mg IM × 2, 24 h apart**; dexamethasone **6 mg IM q12h × 4**; rescue if **≥7 days** and **<34 wk**
- MgSO₄ neuroprotection **6 g bolus + 2 g/h**; CP **1.9% vs 3.5%**; **NNT 63**; avoid tocolytic use **>5–7 days**
- PPROM antibiotics: IV ampicillin + erythromycin **48 h**, then oral amoxicillin + erythromycin **5 days**; latency at 7 d **50% vs 25%**

- Chorioamnionitis (ACOG): $\geq 39.0^{\circ}\text{C}$, or $38.0\text{--}38.9^{\circ}\text{C}$ + risk factor; pulmonary hypoplasia up to **30%** if PPROM <24 wk
- Tocolysis buys up to **48 h**; terbutaline **0.25 mg SC** for tachysystole; indomethacin ≤ 48 h